Code No: H10016/N/AICTE

FACULTY OF ENGINEERING

B.E. (ECE/ETE/M/P/AE) / (CS/AI&DS/AI&ML/IoT/IT) II - Semester (AICTE) (Backlog) (New) Examination, January 2026

Subject: Engineering Chemistry/ Chemistry

Time: 3 Hours Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.
(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.
(iii) Missing data, if any, may be suitably assumed.

1. a) Write the cell notation of calomel electrode.
   b) Give the specification of potable water.
   c) Write short notes on condensation polymers by giving one example.
   d) Give the requirements of a good fuel.
   e) Write short notes on application of composites.
   f) Explain water line corrosion.
   g) Write the reaction involved in Methanol – Oxygen fuel cell.

2. a) How do you determine the pH of an unknown solution using quinhydrone electrode.
   b) Describe the construction of Zinc – Carbon battery with a neat diagram.

3. a) Discuss the water softening method by ion-exchange method.
   b) Define electro chemical corrosion and explain the mechanism.

4. a) Write the differences between Thermoplastics and Thermosetting.
   b) Explain the mechanism of conduction in poly acetylene by any method.

5. a) Define octane and cetane numbers and give their significance.
   b) Give the composition of LPG and CNG.

6. a) Give the concept and principles of Green Chemistry.
   b) Give the properties and significance of bio-diesel.

7. a) Explain the procedure for the determination of hardness of water by EDTA method.
   b) A coal has the following composition by weight: C = 90% ; O = 3.0%; S = 0.5%; N = 0.5% and ash = 2.5%. Net calorific value of the coal was found to be 8,490.5 Kcal/mole. Calculate the percentage of Hydrogen and higher calorific value of coal.

**** End ****

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